

# SEOSAW dataset manual

## v3.1

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## 1 Introduction

This manual provides information on the structure of the SEOSAW dataset, and how to contribute data to the dataset.

For further guidance on collecting SEOSAW data in the field, you should consult the other [SEOSAW field manuals](#).

## 2 Description of the SEOSAW dataset

This section gives a brief overview of the contents of the latest iteration of the SEOSAW dataset, which is available for research purposes by approval from the SEOSAW network. At the time of writing the latest version is v3.1.

The SEOSAW dataset is built around two key tables:

- `plots.csv` - Plot level data
- `stems.csv` - Stem level data

Each row in the plots table represents a single census in a single plot. Each row in the stems table represents a single stem measurement. Note that a single tree may contain many stems and each stem may be measured multiple times over multiple censuses. See [section 4](#) for a breakdown of columns found in the dataset tables. All data objects are linked by the `plot_id` column, which serves as the primary key for the dataset.

There are also some auxiliary data objects in the SEOSAW data repository:

- `polys.shp` - Polygons for each plot
- `small_stems.csv` - Small stem measurements (below minimum diameter threshold)
- `cwd.csv` - Coarse Woody Debris (CWD) data
- `ground_layer` - Ground layer biomass data

There are also some derived datasets:

- `stems_latest.csv` - Stem data subset for the most recent census of each plot
- `stems_multi.csv` - Stem data subset for plots with multiple censuses
- `plots_latest.csv` - Plot data subset for the most recent census of each plot
- `plots_multi.csv` - Plot data subset for plots with multiple censuses
- `plots_subdiv.csv` - Plot data for 1 ha subdivisions of plots >2 ha
- `polys_subdiv.shp` - Polygons for 1 ha subdivisions of plots >2 ha
- `plot_clusters.csv` - Floristic clusters for a subset of the plot data
- `cluster_dom.csv` - Dominant species per cluster

### 2.1 Nesting and subplots

Researchers occasionally stratify their sampling effort across a plot, often according to stem diameter classes, to reduce the work required to survey a plot. This stratification is often referred to as ‘nesting’. See the diagram below for an example of a nested plot. In this example, there are three levels of nesting. Across the whole plot (0.2 ha) stems >10 cm DBH are measured. Within the first level of nesting (N1, 0.1 ha), stems >5 cm DBH are measured, within the second level (N2, 0.05 ha) stems >2 cm DBH, and within the third level (N3, 0.025 ha) stems >1 cm DBH.

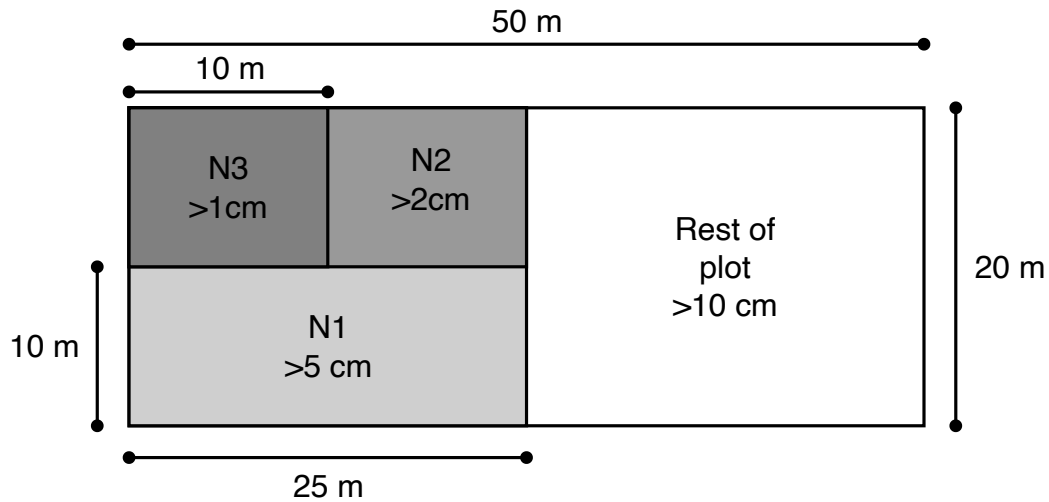


Figure 1: Schematic diagram of a nested plot with three levels of nesting.

There are a number of columns in `plots.csv` which record nesting, prefixed with either `nested_`, `n1`, `n2` or `n3`, for different levels of nesting.

If plots are nested, with variation in sampling effort for trees of different sizes, the Fractional Probability of Inclusion in the `fpc` column of the stem table describes the sampling effort for each stem. In the example, stems >10 cm DBH are measured across the whole plot, so their `fpc` is 1.0. Stems >5 cm DBH are measured in an area 20x25 m (0.1 ha), which is half the total plot area, so their `fpc` is 0.5. Stems >2cm are measured over 0.05 ha, which is a quarter of the total plot area, so their `fpc` is 0.25, and so on. At present, the SEOSAW dataset only allows up to three levels of nesting, but this may change in the future.

It is important to distinguish between nesting and subplots. Nested plots, as demonstrated above, are smaller parcels within a plot which denote variation in sampling effort. In contrast, subplots are merely administrative sub-divisions of a plot which can help with relocating stems for future censuses, for example. Sampling effort should be the same across all subplots. Below is an example of a plot with four 20x25 m subplots.

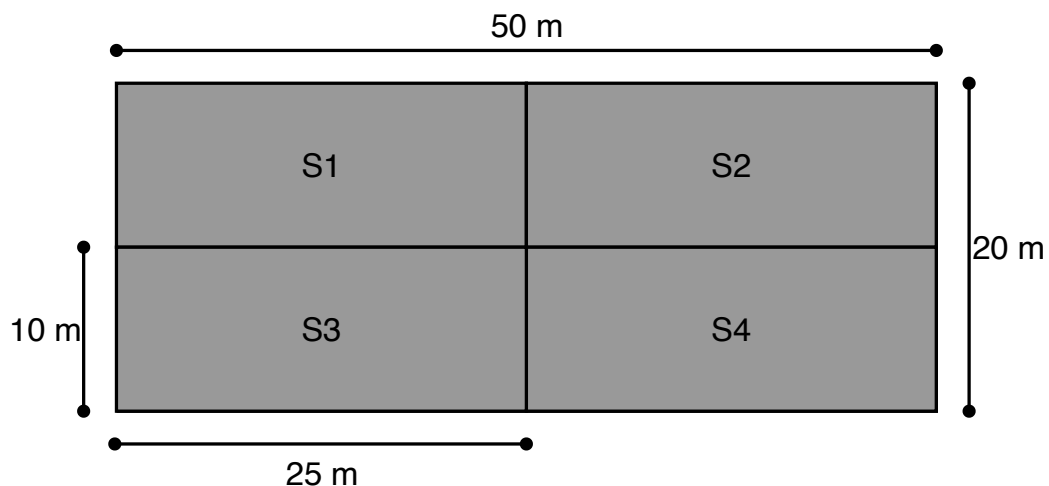


Figure 2: Schematic diagram of a plot with subplots, each occupying 25% of the total plot space.

It is assumed that all subplots have the same dimensions and that subplots are not nested. Columns prefixed with `subplot_` describe subplots.

## 2.2 Stems or trees

The ideal data for the SEOSAW dataset is organised so that there is one measurement for each stem during each census, with stems grouped into trees and a permanent tag to identify each stem. Understandably however, not all data matches this ideal, and is dependent on the original data collection protocol. There are a few columns in the plots table which help describe how stem data is organised:

- `tags` - Do stems have permanent tags which allow for remeasurement?
- `tag_tree_stem` - Are tags applied to each tree, or each stem?
- `meas_tree_stem` - Are measurements recorded at the tree or stem level?
- `tree_diff` - If `meas_tree_stem == 'stem'`, were stem measurements grouped into separate trees?

## 2.3 Permanent or one-off plot

A permanent plot in the SEOSAW dataset is defined as one where trees and new recruits can be measured at a later date. The boundaries of a permanent plot must be clearly identifiable, i.e. with corner posts for a rectangular plot or a central marker for a circular plot.

Related, in many permanent plots it is possible to track the growth and mortality of individual stems between multiple censuses within a plot. Normally this is achieved through the use of numbered tags, but can alternatively be achieved with a highly accurate grid coordinate system, a differential GPS, or other tagging methods. The `stem_timeline_cont` column records whether the current census can be successfully stem-matched with the previous census. The `abandoned` column records if a plot has been abandoned, i.e. will not receive any more censuses in the future.

One-off plots are those where only a single census has been made, with no way of exactly identifying plot boundaries or tree identity in the future.

Both permanent and one-off plots are accepted by SEOSAW. The `permanent` column in the plots table notes whether a plot is permanent or not.

## 2.4 Subdivided large plots

`plots_subdiv.csv` and `polys_subdiv.shp` hold data on 1 ha square subdivisions of large plots. Large plots are subdivided to make them more readily comparable with other 1 ha plots in the dataset, and to make them easier to work with when conducting spatial analyses. Plots are only subdivided if a number of conditions are met:

- Plot is  $\geq 2$  ha
- Plot is rectangular
- All plot edges are divisible by 100 m

Additionally, stems are only assigned to a subdivision if they have grid coordinates in `x_grid` and `y_grid`. If your stems are located instead by latitude/longitude coordinates, consider using the `gridCoordGen()` function to approximate grid coordinates originating from the southwest corner of the plot. Subdividing plots is performed by `largePlotSplit()` in the `{seosawr}` R package.

The `subdiv` column in the plots table shows whether a plot has been subdivided. The `subdiv_id` column in the stems table shows which subdivision a stem has been assigned to. `subdiv_id` matches in `plots_subdiv.csv`

where subdivision-level calculated values (e.g. `agb_ha`) are stored. The calculated values are calculated for each census per plot, like `plots.csv`. If you want only the most recent census for each plot, use `subsetLatest()`. Subdivision IDs are assigned by appending `_S[1-n]` to the plot ID.

## 2.5 Floristic clusters

A portion of the SEOSAW dataset has been clustered based on Ward’s hierarchical clustering method. Data is clustered based on stem abundances per genus in the most recent census per plot. Plots with fewer than 14 stems were excluded from the analysis. Plots with exotic species were also excluded. Data on which cluster each plot is assigned to is found in `plot_clusters.csv` with the dominant species per cluster found in `cluster_dom.csv`.

## 2.6 Taxonomy

The taxonomic ‘backbone’ of the SEOSAW dataset is the [CJB African plants database](#) with some notable exceptions, which are discussed below. The dataset also uses the `taxize::gnr_resolve()` utility to query multiple databases to provide higher taxonomic rank information. The queried databases are:

- [Open Tree of Life Reference Taxonomy](#)
- [The International Plant Names Index](#)
- [Tropicos - Missouri Botanical Garden](#)
- [GBIF Backbone Taxonomy](#)
- [NCBI](#)

The breakup of the *Acacia* genus has been well documented by (Kyalangalilwa et al., 2013). SEOSAW uses their list of synonyms to correct ex-*Acacia* species. The CJB database does not currently acknowledge the *Acacia* split, so queries to CJB retain the *Acacia* name.

All *Rhus* species in SEOSAW’s study region belong to *Searsia* and are transferred in this way. All recent phylogenetic results point to this, but the formal taxonomic work to transfer African *Rhus* species is incomplete.

Previously, *Marquesia* species were placed in the *Trillesanthus* genus, however this has since been reverted. SEOSAW corrects *Trillesanthus* to *Marquesia*.

The latest taxonomy states that *Maytenus* is a strictly Neotropical genus, while acknowledging that African *Maytenus* are a “problem”. The latest taxonomic literature doesn’t tell us what to do with African *Maytenus*. SEOSAW defaults to the CJB taxonomy on *Gymnosporia*/*Maytenus*, but this may change in the future.

There are also some miscellaneous synonym corrections which are maintained throughout the canonical SEOSAW dataset. The vast majority of these corrections reflect previously debunked names which still appear in older datasets.

If a species cannot be identified, it is given the “indet” epithet. If only the genus is known, it is recorded as “*Genus* indet” (e.g. *Brachystegia* indet). If only the family is known, it is recorded as “Family indet” (e.g. Fabaceae indet). As a last resort, if only the local name of the species is known, it is recorded in the `species_orig_local` column.

## 2.7 Biomass

Estimates of biomass, specifically above-ground woody biomass (AGB) are stored in the `agb` column in the stems table. Additionally, plot level above-ground woody biomass is stored in the `agb_ha` (AGB per hectare) column

in the plots table. Biomass is estimated in the SEOSAW dataset by default using the `chave14()` function, which uses Equation 7 from (Chave et al., 2014). This allometric equation takes into account species wood density, stem diameter and an environmental stress index (E), derived from the plot location.

## 2.8 Plot polygons

Plot outline polygons are provided in `polys.shp`, for each plot where appropriate location and plot dimensions data is available. For rectangular plots, these polygons are normally created from GPS coordinates taken at each corner and used as vertices when transforming to a closed spatial polygon.

The SEOSAW region is provided as a spatial data object in the `{seosaw}` R package (`data(seosaw_region)`), along with a simplified bounding box of the SEOSAW region (`data(seosaw_bbox)`). There is also a useful outline of all countries in Africa (`data(africa)`).

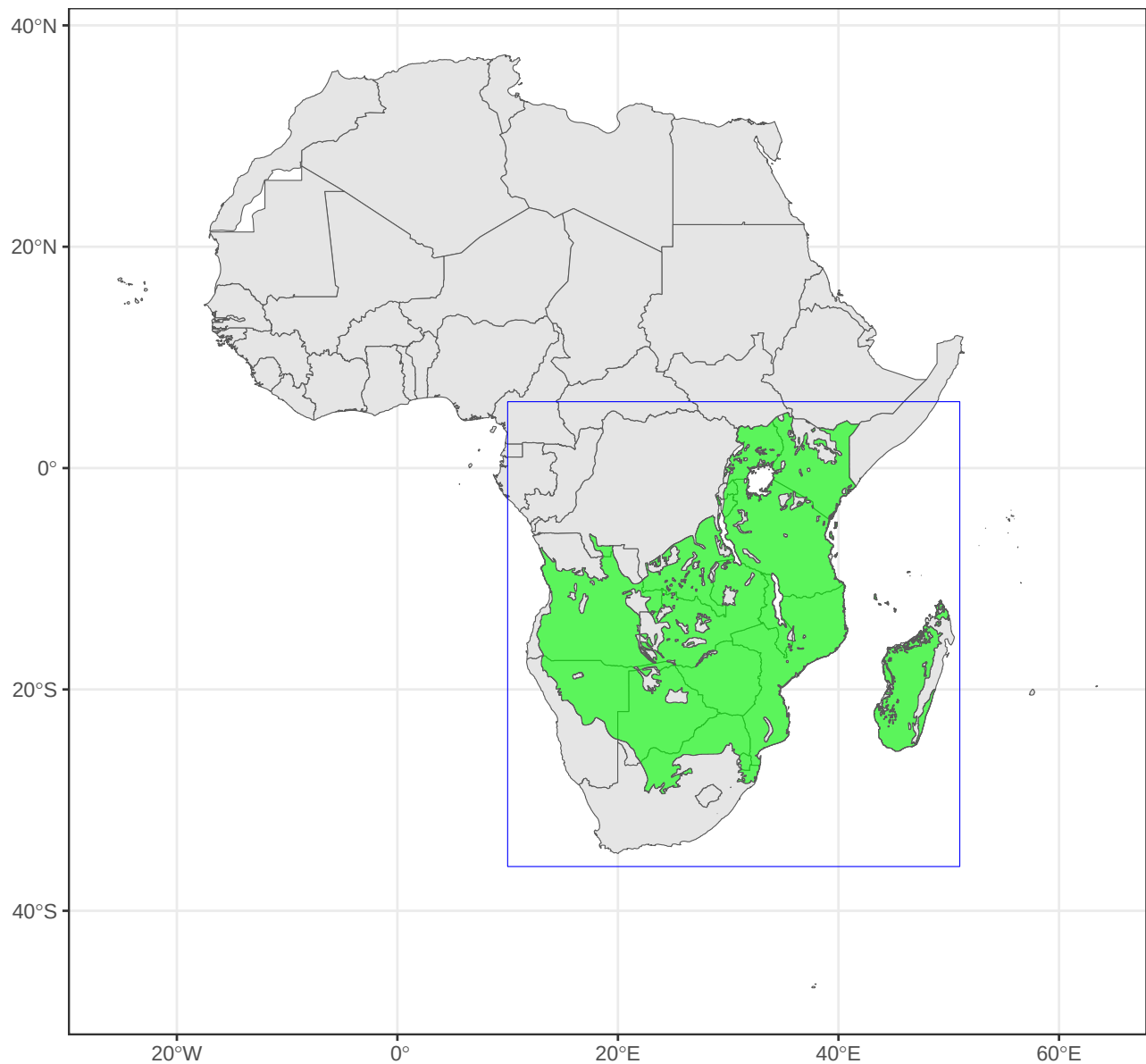


Figure 3: The spatial extent of the SEOSAW region shown in green and the SEOSAW bounding box as a blue rectangle, within Africa.

The SEOSAW region is based largely on White's vegetation map, which defines the locations of savannas and woodlands (White, 1983).

A key issue was defining the north-eastern boundary of the SEOSAW region. It is defined here as the borders of Kenya and Uganda, excluding Ethiopia, Somalia, and South Sudan. While the decision to use country boundaries was largely pragmatic, it is backed up by environmental data. The mountainous Ethiopian highlands on the border between Kenya, Ethiopia and Somalia cause quite a clear change in precipitation, probably due to them blocking dry Arabian air moving further southwest.

The north-western boundary largely follows the rift valley lakes, with everything to the west of them being excluded. Some areas to the east of the lakes are also excluded as they are classed by White's vegetation map as

Congo Forest Transitions and Mosaics.

The south-east of the region is further reduced from White's vegetation map according to the Terrestrial Ecoregions of the World (Dinerstein et al., 2017). White classes some areas as woodland or forest in this area, but for our purposes they are considered too temperate, and are thus excluded from the SEOSAW region.

## 2.9 Dataset caveats

The SEOSAW dataset is aggregated from many individual datasets contributed by researchers around the world. While effort has been made to ensure that datasets are compatible with each other, particularly with respect to stem data like stem `diam` and `height`, some pieces of metadata should not be compared among datasets. For example, `canopy_cover` is collected with hemispherical photos, different types of densitometer, aerial photos and many other methods. Due to the subjectivity of all these methods, it is not recommended to use this data as a variable in statistical analysis when considering multiple datasets; it should be used only as metadata.

SEOSAW takes the stance that metadata should be left blank if it is not known, rather than guessed at. This means that many metadata columns consist mainly of `NA` values. For example, `charcoal_harvesting` is left mostly blank because most data originators did not record whether wood collection for charcoal harvesting had occurred in the plot. This also applies to stem data. If for example, the data originator did not record whether stems were located on termite mounds, `on_termites` is left as `NA`, even if it is more likely that most stems were not located on termite mounds.

## 3 Contributing data

Please submit tabular data as `.csv` (comma-separated values) files if possible. If working in Microsoft Excel or a similar spreadsheet program, there is usually an option to export as `.csv`. `.csv` files are preferred as they are simple and store data exactly as entered. `.xls` or `.xlsx` files are also acceptable.

Spatial data such as plot outlines are preferred as shapefiles (`.shp`), though plot corner or plot centre locations as latitude-longitude coordinates, stored in a `.csv` file are also acceptable. Make sure to describe the Coordinate Reference System (CRS) used for coordinates, WGS84 Latitude-Longitude (EPSG: 4326) is preferred.

Please provide a description of the contents of each file in a separate document. For tabular data, if not using the SEOSAW column definitions, provide a description of the contents of each column. For example, to describe the contents of a column in the stems table containing stem diameter measurements:

`S diam` - Stem diameters, measured in centimetres to one decimal place (i.e. to the nearest mm).

Provide a description of the methodology used to collect the data in a separate document, especially any plot nested-ness or stratified sampling, as this will determine whether any corrections should be made when calculating plot-level statistics like aboveground biomass.

### 3.1 Vital data

There are many columns which can be filled by data contributors in the stem and plot tables, and you should try to provide data for as many as possible.

Below is a list of the columns which are absolutely necessary for minimum data quality. Data submitted without all of these columns cannot be added to the SEOSAW dataset. The column names as they appear in the template data tables are shown in brackets:

Plots table:

- `plot_name` - Plot name, for matching with plot names in stems table
- `longitude_of_centre`, `latitude_of_centre` - Plot longitude and latitude coordinates
- `plot_shape`, `plot_length`, `plot_width`, or `plot_radius` - Plot shape and dimensions
- `prinv` - The Primary Investigator of the data
- `nested_1`, `n1_*`, etc. - Information on plot nested-ness, including nested plot dimensions and stem size thresholds.

Stems table:

- `plot_name` - Plot name, for matching with plot names in plots table
- `species_orig_binom` or `species_orig_local` - Tree species name
- `diam` - Tree diameter
- `pom` - Height of diameter measurement
- `fpc` - Fractional Probability of Inclusion. If plots are nested with varying sampling effort within plots for stems of different size, the probability of a stem this size occurring in the whole plot, based on the proportional area of the nested plot. If plots are not nested, `fpc` = 1.

If only a few entries from each of these columns is missing due to mistakes during fieldwork, don't worry, you data is probably still usable, but try to fill as many empty values as possible.

## 3.2 Additional data

Please provide *any* metadata you think might be even slightly useful to gain a more detailed understanding of your plots. For example, if you have meteorological data for a nearby weather station, herbivore dung counts, photos of the plots, land use history timelines, soil core profiles, or fire management notes. As with the plot and stem data, try to provide a detailed description of the data, the methodology used to collect the data, and any caveats.

## 4 Dataset table columns

This section provides a comprehensive description of every column found in the SEOSAW dataset, across the plots table, stems table, small stems table, coarse woody debris table, and ground layer table.

Table 1: Column names for plots table.

| Column         | Class | Description                                                                                                                                                                                                                                                                                                                                                                                           | Units | Example                               |
|----------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------|
| plot_id        | chr   | Standardised method of identifying plots. A three letter code, followed by a number referring to the particular plot. Three letter codes are normally, but not always, created like this: Angola (A), in Bicuar National Park (B), by John Godlee (G). The three letter code should be conserved within a dataset, i.e. a group of plots owned by a particular researcher within a particular locale. | NA    | SSM_1                                 |
| census_date    | chr   | The date of the census. This can be formatted as either YYYY, YYYY-MM, or YYYY-MM-DD                                                                                                                                                                                                                                                                                                                  | date  | 2008                                  |
| plot_cluster   | chr   | If plots occur in a spatial cluster, a single 'plot_id' used to identify the cluster.                                                                                                                                                                                                                                                                                                                 | NA    | ABG_1                                 |
| plot_name      | chr   | Plot name as designated by those who originally collected the data. 1, Wedza_01, MHR-04b are all valid. No spaces please.                                                                                                                                                                                                                                                                             | NA    | Wedza_01                              |
| country_iso3   | fct   | ISO 3166-1 alpha-3 code to refer to the country in which the plot is situated. Only countries within Africa. See 'data(africa_iso3)' for available country codes. use countrycode::countrycode() to convert a country name to an ISO3 code                                                                                                                                                            | NA    | AGO                                   |
| prinv          | chr   | Name of the Principal Investigator. This can be an organisation, a person's name, a surname, initials, etc.                                                                                                                                                                                                                                                                                           | NA    | Godlee J. L.                          |
| other_authors  | chr   | Names of other authors who contributed to the collection of the dataset, separated by semi-colons                                                                                                                                                                                                                                                                                                     | NA    | Godlee J. L.; Maiato F.; Dexter K. G. |
| plot_shape     | fct   | Shape of the plot. Possible values are: "rectangle", "circle" or "irregular". Note that a square is a special case of "rectangle".                                                                                                                                                                                                                                                                    | NA    | rectangle                             |
| plot_width     | int   | If plot_shape == "rectangle", the shorter side length, in metres to the nearest metre                                                                                                                                                                                                                                                                                                                 | m     | 100                                   |
| plot_length    | int   | If plot_shape == "rectangle", the longer side length, in metres to the nearest metre                                                                                                                                                                                                                                                                                                                  | m     | 50                                    |
| plot_radius    | int   | If plot_shape == "circle", the radius of the plot, measured in metres to the nearest metre                                                                                                                                                                                                                                                                                                            | m     | 25                                    |
| subplot_n      | int   | Number of subplots within the larger plot. Subplots (as opposed to nested plots) have the same sampling effort as the entire plot, generally used for ease of relocating trees.                                                                                                                                                                                                                       | n     | 10                                    |
| subplot_shape  | fct   | Shape of the subplots. Possible values are the same as plot_shape                                                                                                                                                                                                                                                                                                                                     | NA    | circle                                |
| subplot_length | int   | If subplot_shape == "rectangle", the longest side length, in metres to the nearest metre                                                                                                                                                                                                                                                                                                              | m     | 50                                    |
| subplot_width  | int   | If subplot_shape == "rectangle", the shorter side length, in metres to the nearest metre                                                                                                                                                                                                                                                                                                              | m     | 10                                    |

Table 1: Column names for plots table. *(continued)*

| Column               | Class | Description                                                                                                                                                                               | Units | Example   |
|----------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|
| subplot_radius       | int   | If subplot_shape == "circle", the radius of the plot, measured in metres to the nearest metre                                                                                             | m     | 10        |
| min_diam_thresh      | int   | Minimum stem diameter value considered within this plot, in centimetres to the nearest centimetre. If sampling is nested this should be the minimum value outside of all nested subplots. | cm    | 5         |
| pom_default          | num   | The default height at which diameter is measured, in metres to one decimal place (nearest 10 cm)                                                                                          | m     | 1.3       |
| min_height_thresh    | num   | Minimum stem height value considered within this plot, in metres to the nearest 10 centimetres. If sampling is nested this should be the maximum value outside of all nested subplots.    | m     | 1.5       |
| nested_1             | log   | Does the plot have at least one level of nested sampling?                                                                                                                                 | NA    | TRUE      |
| n1_min_diam_thresh   | int   | If nested_1 == TRUE, the minimum stem diameter value considered within the first nesting level, in centimetres to the nearest centimetre.                                                 | cm    | 10        |
| n1_min_height_thresh | num   | If nested_1 == TRUE, the minimum stem height value considered within the first nesting level, in metres to the nearest metre.                                                             | m     | 20        |
| n1_plot_shape        | fct   | If nested_1 == TRUE, the shape of the subplot, same values as plot_shape                                                                                                                  | NA    | circle    |
| n1_plot_width        | int   | If nested_1 == TRUE & n1_plot_shape == "rectangle" the shorter side length, in metres to the nearest metre                                                                                | m     | 20        |
| n1_plot_length       | int   | If nested_1 == TRUE & n1_plot_shape == "rectangle" the longer side length, in metres to the nearest metre                                                                                 | m     | 10        |
| n1_plot_radius       | int   | If nested_1 == TRUE & n1_plot_shape == "circle" the radius of the subplot, in metres to the nearest metre                                                                                 | m     | 14        |
| nested_2             | log   | Does the plot have at least two levels of nested sampling?                                                                                                                                | NA    | TRUE      |
| n2_min_diam_thresh   | int   | If nested_2 == TRUE, the minimum stem diameter value considered within the second nesting level, in centimetres to the nearest centimetre.                                                | cm    | 5         |
| n2_min_height_thresh | num   | If nested_2 == TRUE, the minimum stem height value considered within the second nesting level, in metres to the nearest metre.                                                            | m     | 2.5       |
| n2_plot_shape        | fct   | If nested_2 == TRUE, the shape of the subplot, same allowed values as plot_shape                                                                                                          | NA    | rectangle |
| n2_plot_width        | int   | If nested_2 == TRUE & n2_plot_shape == "rectangle" the shorter side length, in metres to the nearest metre                                                                                | m     | 30        |
| n2_plot_length       | int   | If nested_2 == TRUE & n2_plot_shape == "rectangle" the longer side length, in metres to the nearest metre                                                                                 | m     | 20        |

Table 1: Column names for plots table. (*continued*)

| Column                      | Class | Description                                                                                                                               | Units | Example   |
|-----------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|
| n2_plot_radius              | int   | If nested_2 == TRUE & n2_plot_shape == "circle" the radius of the subplot, in metres to the nearest metre                                 | m     | 5         |
| nested_3                    | log   | Does the plot have at least three levels of nested sampling?                                                                              | NA    | TRUE      |
| n3_min_diam_thresh          | int   | If nested_3 == TRUE, the minimum stem diameter value considered within the third nesting level, in centimetres to the nearest centimetre. | cm    | 5         |
| n3_min_height_thresh        | num   | If nested_3 == TRUE, the minimum stem height value considered within the third nesting level, in metres to the nearest metre.             | m     | 2.5       |
| n3_plot_shape               | fct   | If nested_3 == TRUE, the shape of the subplot, same allowed values as plot_shape                                                          | NA    | rectangle |
| n3_plot_width               | int   | If nested_3 == TRUE & n3_plot_shape == "rectangle" the shorter side length, in metres to the nearest metre                                | m     | 30        |
| n3_plot_length              | int   | If nested_3 == TRUE & n3_plot_shape == "rectangle" the longer side length, in metres to the nearest metre                                 | m     | 20        |
| n3_plot_radius              | int   | If nested_3 == TRUE & n3_plot_shape == "circle" the radius of the subplot, in metres to the nearest metre                                 | m     | 5         |
| lianas_sampled              | log   | Were lianas sampled? I.e. woody stems that are not self-supporting and use other trees to grow towards the canopy                         | NA    | TRUE      |
| soil_sampled                | log   | Were soil samples collected?                                                                                                              | NA    | FALSE     |
| ground_layer_sampled        | log   | Was the ground layer sampled for grasses, forbs, shrubs, or small trees that did not fit the size thresholds for the plot?                | NA    | TRUE      |
| dead_stems_sampled          | log   | Were dead stems sampled in this plot?                                                                                                     | NA    | FALSE     |
| stumps_sampled              | log   | Were tree stumps sampled?                                                                                                                 | NA    | TRUE      |
| coarse_woody_debris_sampled | log   | Was coarse woody debris sampled?                                                                                                          | NA    | FALSE     |
| small_stems_sampled         | log   | Were woody stems smaller than the minimum size thresholds counted?                                                                        | NA    | TRUE      |
| tags                        | log   | Are stems or trees within the plot tagged or otherwise identifiable between censuses?                                                     | NA    | FALSE     |
| permanent                   | log   | Is the plot a permanent plot (TRUE) or a one-off census (FALSE)? Could the plot be re-measured in the future?                             | NA    | TRUE      |
| meas_tree_stem              | fct   | Are measurements taken for each "tree", or each "stem"?                                                                                   | NA    | tree      |
| tag_tree_stem               | fct   | Are tag IDs at the "tree" or "stem" level?                                                                                                | NA    | tree      |
| tree_diff                   | log   | Are stems grouped into trees, where trees have multiple stems?                                                                            | NA    | TRUE      |

Table 1: Column names for plots table. *(continued)*

| Column               | Class | Description                                                                                                                                                                                                                                                                                     | Units | Example                                                        |
|----------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------|
| abandoned            | log   | Is this the last census and the plot is now abandoned? Common reasons for abandonment: Clear-felling or nearly all trees removed. Often combined with tag loss and irreconcilable damage to the plot which would make it hard to match the stems with a previous census.                        | NA    | TRUE                                                           |
| stem_timeline_cont   | log   | If tags == TRUE and permanent == TRUE, is it impossible to link stems between this census and any previous censuses?                                                                                                                                                                            | NA    | FALSE                                                          |
| sampling_notes       | chr   | Extra notes relating to the sampling process                                                                                                                                                                                                                                                    | NA    | Only stumps greater than 2 m tall were sampled                 |
| sampling_design_code | chr   | File name for document containing the description of your methods/protocol for data collection, provided to SEOSAW, or a reference to such a document. If not documented then NA                                                                                                                | NA    | SSAW_V1.gdoc                                                   |
| plot_notes           | chr   | Include any long form comments about the plot here, if they don't fit into other columns.                                                                                                                                                                                                       | NA    | It looks like this plot is a rubbish dump for household waste. |
| longitude_of_centre  | num   | Longitude of the centre of gravity of the plot. If you have GPS points of the plot corners, send those and we will calculate this. Measured in decimal degrees, WGS84 EPSG:4326                                                                                                                 | deg   | -50.425622                                                     |
| latitude_of_centre   | num   | Latitude of the centre of gravity of the plot. If you have GPS points of the plot corners, send those and we will calculate this. Measured in decimal degrees, WGS84 EPSG:4326                                                                                                                  | deg   | 30.446453                                                      |
| elevation            | num   | Elevation of the plot. Measured in metres above International Mean sea level to 1 decimal place (nearest 10 cm). In order of preference: 1) Average of GPS readings across the plot, 2) a single GPS reading, 3) Manual lookup on topographic map, 4) NA. We can query SRTM for missing values. | m asl | 1234                                                           |
| slope_method         | fct   | Method used to determine plot slope. Allowed values: "rangefinder" if a laser or sonic rangefinder was used, "trig" if slope was estimated with tape measures and-or a clinometer, "eye" if estimated by eye, "other" if another method.                                                        | NA    | trig                                                           |
| slope_method_notes   | chr   | Extra notes relating to the slope method                                                                                                                                                                                                                                                        | NA    | 20 points taken with 20 m minimum distance, randomly located   |
| slope                | num   | Plot slope, measured in degrees from 0 (flat) to 90 (vertical)                                                                                                                                                                                                                                  | deg   | 4.4                                                            |

Table 1: Column names for plots table. (*continued*)

| Column                    | Class | Description                                                                                                                                                                                                         | Units      | Example                                                                                |
|---------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------|
| aspect                    | num   | Aspect of the plot, i.e. which direction is the slope facing, in degrees (0-360)                                                                                                                                    | deg        | 54                                                                                     |
| polygons                  | log   | Polygon data of the plot shape provided by the data originator?                                                                                                                                                     | NA         | TRUE                                                                                   |
| local_allometry           | fct   | Most appropriate aboveground woody biomass allometry for the plot. In order of preference: (1) DOI, (2) reference, (3) other URL, (4) shorthand allometry name                                                      | NA         | ryan11                                                                                 |
| allometry_notes           | chr   | Extra notes relating to the appropriate local allometry                                                                                                                                                             | NA         | ryan11 should be used for everything except the eucalyptus trees present in plot ABC_5 |
| catenal_position          | fct   | Where on the catena is plot? Possible values are:"ridge_crest", "concave_slope", "midslope", "convex_slope", "valley_floor", "flat"                                                                                 | NA         | midslope                                                                               |
| termites                  | int   | If termites were counted, the number of termite mounds found on the plot                                                                                                                                            | n          | 34                                                                                     |
| canopy_cover              | prop  | The proportion of the ground (periscope, aerial photo, canopy dimensions), or sky hemisphere (hemi-photo, mirror densitometer), covered by a tree canopy. Measured as a proportion between 0 and 1, e.g. 60% = 0.6  | proportion | 0.28                                                                                   |
| canopy_cover_method       | fct   | If canopy cover was estimated, what method was used? Possible values are: "periscope_densitometer", "mirror_densitometer", "hemispherical_photo", "aerial_photo", "canopy_dimensions", "other"                      | NA         | hemispherical_photo                                                                    |
| canopy_cover_method_notes | chr   | Extra notes relating to the canopy cover method                                                                                                                                                                     | NA         | 40 hemispherical photos in a regular grid with 20 m spacing. Analysed with Hemiphot.   |
| xy_method                 | fct   | Method used to estimate stem location within the plot, either grid or GPS coordinates. Possible values: "tape", "gps", "dgps", "angle" (angular direction and distance from plot centre in circular plot), "other". | NA         | tape                                                                                   |
| manipulation_experiment   | log   | Has any aspect of the plot's land use or environment been manipulated by researchers?                                                                                                                               | NA         | FALSE                                                                                  |

Table 1: Column names for plots table. *(continued)*

| Column                         | Class | Description                                                                                                                                                                                                         | Units | Example                               |
|--------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------|
| experiment_ref                 | chr   | If manipulation_experiment == TRUE, a link, citation, or contact details relating to the experimental manipulation of this plot. In order of preference: (1) DOI, (2) reference, (3) other URL, (4) contact details | NA    | 10.1890/06-1664                       |
| manipulation_experiment_notes  | chr   | If manipulation_experiment == TRUE, notes relating to the experimental treatment applied to the plot                                                                                                                | NA    | Tilling of soil 15-20 cm throughout   |
| pa                             | log   | Is the plot in a protected area?                                                                                                                                                                                    | NA    | TRUE                                  |
| pa_name                        | chr   | If pa == TRUE, the name of the protected area. If the protected area is listed by the WDPA, please use the name provided by the WDPA                                                                                | NA    | Bicuar National Park                  |
| pa_type                        | chr   | If pa == TRUE, the type of protected area. If the protected area is listed by the WDPA, use their type classification.                                                                                              | NA    | Forest reserve                        |
| wdpa_id                        | chr   | If pa == TRUE and the protected area is listed by the WDPA, the WDPA ID value for that protected area. See <a href="https://www.protectedplanet.net">https://www.protectedplanet.net</a>                            | NA    | 370                                   |
| local_community_rights         | chr   | Explain the rights of the local community in using the plot for resource extraction, grazing, farming etc.                                                                                                          | NA    | Dead wood collection by local village |
| other_user_rights              | chr   | Explain the rights of any others in using the plot for resource extraction, grazing, farming etc.                                                                                                                   | NA    | Yearly game drive in early May        |
| timber_harvesting              | log   | Is there evidence of tree harvesting for timber products, e.g. planks?                                                                                                                                              | NA    | TRUE                                  |
| high_graded_100_years          | log   | Is it likely that high quality timber trees were removed from the plot the last 100 years?                                                                                                                          | NA    | FALSE                                 |
| clear_cut                      | log   | Has any part of the plot ever been clear cut (all trees removed)?                                                                                                                                                   | NA    | TRUE                                  |
| mining                         | log   | Has any part of the plot ever been used for mining?                                                                                                                                                                 | NA    | FALSE                                 |
| fuel_wood_harvesting           | log   | Is there evidence of fuel wood collection, excluding charcoal production?                                                                                                                                           | NA    | TRUE                                  |
| charcoal_harvesting            | log   | Is there evidence of trees being harvested for charcoal production?                                                                                                                                                 | NA    | FALSE                                 |
| other_woody_product_harvesting | log   | Is there evidence of people harvesting woody material for any other use, e.g. string, beehives, canoes?                                                                                                             | NA    | TRUE                                  |
| ntfp_harvesting                | log   | Is there evidence of people harvesting non-wood tree products, e.g. mushrooms honey, fruit etc.?                                                                                                                    | NA    | FALSE                                 |
| farmed_30_years                | log   | Is it likely that the plot was used for farming crops in the last 30 years?                                                                                                                                         | NA    | TRUE                                  |
| livestock_30_years             | log   | Is it likely that the plot was used for grazing livestock in the last 30 years?                                                                                                                                     | NA    | FALSE                                 |
| land_use_notes                 | chr   | Extra notes relating to land use in the plot                                                                                                                                                                        | NA    | FALSE                                 |

Table 1: Column names for plots table. *(continued)*

| Column                   | Class | Description                                                                                                                                                                                                                          | Units              | Example                                                                 |
|--------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------|
| fire_exclusion           | log   | Is there a fire exclusion experiment in this plot? If TRUE, manipulation_experiment should also = TRUE.                                                                                                                              | NA                 | TRUE                                                                    |
| fire_exclusion_success   | fct   | If fire_exclusion == TRUE, to what extent were fires successfully excluded from the plot? Possible values: "all", "most", "few"                                                                                                      | NA                 | all                                                                     |
| fire_treatment           | log   | Is or was there a fire treatment experiment in this plot, e.g. controlled fires every two years? If TRUE, manipulation_experiment should also = TRUE.                                                                                | NA                 | FALSE                                                                   |
| fire_treatment_season    | fct   | If fire_treatment == TRUE, the season in which fires are set. Possible values: "early_dry_season", "late_dry_season", "early_wet_season", "late_wet_season"                                                                          | NA                 | early_dry_season                                                        |
| fire_return_interval     | int   | Number of years between fires. If there has been some sort of fire control experiment, the treatment fire return interval measured in years, e.g. 1 for annual burning. If no treatment, the estimated mean interval between fires.  | year               | 5                                                                       |
| fire_last_date           | chr   | Date of the last fire.                                                                                                                                                                                                               | date               | 2022                                                                    |
| fire_cause               | chr   | What commonly causes fires in the plot? E.g. Management, accidental, lightning, agriculture.                                                                                                                                         | NA                 | Adjacent field fires                                                    |
| fire_fuel                | chr   | What do fires on this plot typically burn? E.g. Canopy, understorey, litter                                                                                                                                                          | NA                 | Tree canopies                                                           |
| fire_notes               | chr   | Extra notes relating to fire regime or fire treatment                                                                                                                                                                                | NA                 | Fires are set with drip torches in the late evening, after a heavy rain |
| elephants                | log   | Do elephants frequent the plot                                                                                                                                                                                                       | NA                 | TRUE                                                                    |
| elephant_density         | num   | If elephants are present, at what approximate density do they occur within the area? Measured as elephants per km <sup>2</sup>                                                                                                       | n km <sup>-2</sup> | 5                                                                       |
| elephant_data_source     | chr   | A link, citation, or contact details relating to elephant density estimates. In order of preference: (1) DOI, (2) reference, (3) other URL, (4) contact details                                                                      | NA                 | 10.6088/ijes.002                                                        |
| exclosure                | fct   | Are herbivores excluded from the plot by a fence? If so, what sizes? Possible values are: "extra_large" (>1000 kg), "large" (100-1000 kg), "medium" (30-100 kg), "small" (<30 kg). Multiple values possible separated by semi-colon. | NA                 | extra_large                                                             |
| small_nonsocial_browsers | log   | Do small non-social browsers frequent this plot? e.g. dik-diks, duikers and grysbok, steenbok (these categories are based on Hempson et al 2015). TRUE or FALSE                                                                      | NA                 | TRUE                                                                    |
| large_browsers           | log   | Do large browsers frequent this plot? e.g. giraffe, okapi, kudu                                                                                                                                                                      | NA                 | FALSE                                                                   |

Table 1: Column names for plots table. (*continued*)

| Column                         | Class | Description                                                                                                                                                              | Units | Example                                                                     |
|--------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------|
| medium_sized_social_mixed_diet | log   | Do medium sized social mixed diet herbivores frequent this plot?<br>e.g. gazelle, oryx                                                                                   | NA    | TRUE                                                                        |
| nonruminants_excluding_suids   | log   | Do non-ruminant herbivores frequent this plot? e.g. zebra, hippo, rhino                                                                                                  | NA    | FALSE                                                                       |
| water_dependent_grazers        | log   | Do water-dependent grazers frequent this plot? e.g. roan, sable, bluebuck, waterbuck, lechwe                                                                             | NA    | TRUE                                                                        |
| goat_grazing                   | log   | Do goats feed in this plot?                                                                                                                                              | NA    | FALSE                                                                       |
| sheep_grazing                  | log   | Do sheep feed in this plot?                                                                                                                                              | NA    | TRUE                                                                        |
| cattle_grazing                 | log   | Do domestic cattle feed in the plot?                                                                                                                                     | NA    | FALSE                                                                       |
| herbivore_notes                | chr   | Extra notes relating to herbivores in the plot.                                                                                                                          | NA    | Dik Diks, palankas, Olongo, elephants like to knock over the B. massaiensis |
| cyclone                        | log   | Do cyclones (category 3+) ever occur in this plot?                                                                                                                       | NA    | TRUE                                                                        |
| cyclone_last_date              | chr   | When was the last cyclone?                                                                                                                                               | NA    | 2022                                                                        |
| drought                        | log   | Do major droughts ever occur in this plot?                                                                                                                               | NA    | FALSE                                                                       |
| drought_last_date              | chr   | When was the last major drought?                                                                                                                                         | NA    | 2021                                                                        |
| wind                           | log   | Do wind storms or tornadoes ever occur in this plot?                                                                                                                     | NA    | TRUE                                                                        |
| wind_last_date                 | chr   | When was the last wind storm or tornado?                                                                                                                                 | NA    | 2020                                                                        |
| earthquake                     | log   | Do earthquakes ever occur in this plot?                                                                                                                                  | NA    | FALSE                                                                       |
| earthquake_last_date           | chr   | When was the last earthquake?                                                                                                                                            | NA    | 2019                                                                        |
| landslide                      | log   | Do landslides ever occur in this plot?                                                                                                                                   | NA    | TRUE                                                                        |
| landslide_last_date            | chr   | When was the last landslide?                                                                                                                                             | NA    | 2018                                                                        |
| flood                          | log   | Do major floods ever occur in this plot?                                                                                                                                 | NA    | FALSE                                                                       |
| flood_last_date                | chr   | When was the last major flood?                                                                                                                                           | NA    | 2017                                                                        |
| frost                          | log   | Do major frost events (<-2C for three consecutive nights) ever occur in this plot?                                                                                       | NA    | TRUE                                                                        |
| frost_last_date                | chr   | When was the last major frost event?                                                                                                                                     | NA    | 2016                                                                        |
| data_ref                       | chr   | A link, citation, or contact details relating to the publication of this plot's data. In order of preference: (1) DOI, (2) reference, (3) other URL, (4) contact details | NA    | 10.3390/d12040140                                                           |
| photos                         | chr   | Filenames of photos of the plot, separated by semi-colons                                                                                                                | NA    | DSC_134.jpg;DSC_-135.jpg                                                    |
| date_metadata                  | num   | Year that dataset metadata values, i.e. herbivory, land use, plot location, fire regime etc. were last updated                                                           | NA    | 2020                                                                        |

Table 1: Column names for plots table. *(continued)*

| Column           | Class | Description                                                                                                                     | Units | Example                                         |
|------------------|-------|---------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------|
| file_plots       | chr   | Filepath of the raw plot metadata file provided to SEOSAW. Include multiple files in a semi-colon separated list.               | NA    | archibald_-<br>kruger.xlsx;Kruger_-<br>2024.csv |
| file_stems       | chr   | Filepath of the raw stem data file provided to SEOSAW. Include multiple files in a semi-colon separated list.                   | NA    | Archibald_kruger.csv                            |
| subdiv           | log   | Is the plot subdivided into smaller 1 ha square parcels? If TRUE, plot-level calculated values can be found in plots_subdiv.csv | NA    | TRUE                                            |
| dataset_id       | chr   | The dataset ID, normally defined as the three letter code in plot_id                                                            | NA    | SSM                                             |
| plot_area        | num   | Flat area of the plot, in hectares                                                                                              | ha    | 0.25                                            |
| subplot_area     | num   | Flat area of the subplots, in hectares                                                                                          | ha    | 0.005                                           |
| plot_perimeter   | num   | Perimeter of the plot, in metres                                                                                                | m     | 400                                             |
| diam_max         | num   | Maximum stem diameter found in the plot                                                                                         | cm    | 50.4                                            |
| diam_min         | num   | Minimum stem diameter found in the plot                                                                                         | cm    | 5                                               |
| diam_mean        | num   | Mean stem diameter found in the plot                                                                                            | cm    | 5.6                                             |
| diam_median      | num   | Median stem diameter found in the plot                                                                                          | cm    | 10.3                                            |
| diam_sd          | num   | Standard deviation of stem diameter found in the plot                                                                           | cm    | 1.5                                             |
| diam_ge5_mean    | num   | Mean stem diameter found in the plot, greater than 5 cm diameter                                                                | cm    | 5.4                                             |
| diam_ge5_median  | num   | Median stem diameter found in the plot, greater than 5 cm diameter                                                              | cm    | 10.6                                            |
| diam_ge5_sd      | num   | Standard deviation of stem diameter found in the plot, greater than 5 cm diameter                                               | cm    | 1.5                                             |
| diam_ge10_mean   | num   | Mean stem diameter found in the plot, greater than 10 cm diameter                                                               | cm    | 15.7                                            |
| diam_ge10_median | num   | Median stem diameter found in the plot, greater than 10 cm diameter                                                             | cm    | 15.9                                            |
| diam_ge10_sd     | num   | Standard deviation of stem diameter found in the plot, greater than 10 cm diameter                                              | cm    | 2.5                                             |
| diam_ge20_mean   | num   | Mean stem diameter found in the plot, greater than 20 cm diameter                                                               | cm    | 15.7                                            |
| diam_ge20_median | num   | Median stem diameter found in the plot, greater than 20 cm diameter                                                             | cm    | 15.9                                            |
| diam_ge20_sd     | num   | Standard deviation of stem diameter found in the plot, greater than 20 cm diameter                                              | cm    | 2.5                                             |
| n_diam           | num   | Number of stems for which diameter was measured                                                                                 | n     | 524                                             |
| n_height         | num   | Number of stems for which height was measured                                                                                   | n     | 532                                             |
| height_max       | num   | Maximum stem height found in the plot                                                                                           | m     | 10.5                                            |
| height_min       | num   | Minimum stem height found in the plot                                                                                           | m     | 1.6                                             |
| height_mean      | num   | Mean stem height found in the plot                                                                                              | m     | 1.6                                             |
| height_median    | num   | Median stem height found in the plot                                                                                            | m     | 5.4                                             |
| height_sd        | num   | Standard deviation stem height found in the plot                                                                                | m     | 4.5                                             |

Table 1: Column names for plots table. *(continued)*

| Column             | Class | Description                                                                                                                                                                 | Units      | Example                           |
|--------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------|
| height_ge5_mean    | num   | Mean stem height found in the plot, greater than 5 cm diameter                                                                                                              | m          | 3.5                               |
| height_ge5_median  | num   | Median stem height found in the plot, greater than 5 cm diameter                                                                                                            | m          | 5.4                               |
| height_ge5_sd      | num   | Standard deviation of stem height found in the plot, greater than 5 cm diameter                                                                                             | m          | 1.5                               |
| height_ge10_mean   | num   | Mean stem height found in the plot, greater than 10 cm diameter                                                                                                             | m          | 3.5                               |
| height_ge10_median | num   | Median stem height found in the plot, greater than 10 cm diameter                                                                                                           | m          | 6.6                               |
| height_ge10_sd     | num   | Standard deviation of stem height found in the plot, greater than 10 cm diameter                                                                                            | m          | 1.4                               |
| height_ge20_mean   | num   | Mean stem height found in the plot, greater than 20 cm diameter                                                                                                             | m          | 3.5                               |
| height_ge20_median | num   | Median stem height found in the plot, greater than 20 cm diameter                                                                                                           | m          | 6.6                               |
| height_ge20_sd     | num   | Standard deviation of stem height found in the plot, greater than 20 cm diameter                                                                                            | m          | 1.4                               |
| height_per95       | num   | 95th percentile of stem height, i.e. the height below which 95% of stems can be found                                                                                       | m          | 4.3                               |
| n_stems            | num   | Number of stems in the plot                                                                                                                                                 | n          | 5025                              |
| n_stems_ge5        | num   | Number of stems greater than 5 cm diameter                                                                                                                                  | n          | 5010                              |
| n_stems_ge10       | num   | Number of stems greater than 10 cm diameter                                                                                                                                 | n          | 2040                              |
| n_stems_ge20       | num   | Number of stems greater than 20 cm diameter                                                                                                                                 | n          | 505                               |
| n_stems_ge40       | num   | Number of stems greater than 40 cm diameter                                                                                                                                 | n          | 20                                |
| n_stems_ge60       | num   | Number of stems greater than 60 cm diameter                                                                                                                                 | n          | 1                                 |
| lorey_height       | num   | Lorey height of the plot. Calculated as the sum of basal areas multiplied by height, divided by the total basal area. This variant is adjusted for nested sampling with fpc | NA         | 3.5                               |
| n_determ_genus     | num   | Number of stems where genera successfully identified                                                                                                                        | n          | 20                                |
| n_determ_species   | num   | Number of stems where species epithets successfully identified                                                                                                              | n          | 49                                |
| dom_1              | chr   | Most dominant species in the plot, by stem abundance                                                                                                                        | NA         | Brachystegia spiciformis          |
| dom_2              | chr   | Second most dominant species in the plot, by stem abundance                                                                                                                 | NA         | Ochna pulchra                     |
| dom_3              | chr   | Third most dominant species in the plot, by stem abundance                                                                                                                  | NA         | Burkea africana                   |
| dom_4              | chr   | Fourth most dominant species in the plot, by stem abundance                                                                                                                 | NA         | Indet indet                       |
| dom_5              | chr   | Fifth most dominant species in the plot, by stem abundance                                                                                                                  | NA         | Pseudolachnostylis maprouneifolia |
| prop_1             | num   | Proportional abundance of the most dominant species in the plot                                                                                                             | proportion | 20.4                              |
| prop_2             | num   | Proportional abundance of the second most dominant species in the plot                                                                                                      | proportion | 14.5                              |
| prop_3             | num   | Proportional abundance of the third most dominant species in the plot                                                                                                       | proportion | 12.7                              |

Table 1: Column names for plots table. *(continued)*

| Column           | Class | Description                                                                                                                                    | Units                           | Example |
|------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------|
| prop_4           | num   | Proportional abundance of the fourth most dominant species in the plot                                                                         | proportion                      | 8.4     |
| prop_5           | num   | Proportional abundance of the fifth most dominant species in the plot                                                                          | proportion                      | 3.6     |
| richness         | num   | Species richness of plot                                                                                                                       | n                               | 53      |
| rarefy           | num   | Rarefied species richness at 25 individuals                                                                                                    | NA                              | 42.65   |
| shannon          | num   | Shannon diversity index of plot                                                                                                                | NA                              | 4.3     |
| simpson          | num   | Simpson diversity index of plot                                                                                                                | NA                              | 2.5     |
| invsimpson       | num   | Inverse Simpson diversity index of plot                                                                                                        | NA                              | 2.6     |
| evenness         | num   | Abundance evenness of plot                                                                                                                     | NA                              | 2.6     |
| stocking_ha      | num   | Number of stems per hectare, adjusted for nested sampling with fpc                                                                             | n ha <sup>-1</sup>              | 50.2    |
| stocking_ge5_ha  | num   | Number of stems greater than 5 cm diameter per hectare, adjusted for nested sampling with fpc                                                  | n ha <sup>-1</sup>              | 45.5    |
| stocking_ge10_ha | num   | Number of stems greater than 10 cm diameter per hectare, adjusted for nested sampling with fpc                                                 | n ha <sup>-1</sup>              | 32.5    |
| stocking_ge20_ha | num   | Number of stems greater than 20 cm diameter per hectare, adjusted for nested sampling with fpc                                                 | n ha <sup>-1</sup>              | 32.5    |
| ba_ha            | num   | Basal area per hectare, adjusted for nested sampling with fpc                                                                                  | m <sup>2</sup> ha <sup>-1</sup> | 3.6     |
| agb_ha           | num   | Aboveground woody biomass per hectare, for living stems only (stem_status == "a"   is.na(stem_status)), adjusted for nested sampling with fpc  | t DM ha <sup>-1</sup>           | 42.6    |
| agb_ge5_ha       | num   | Aboveground woody biomass per hectare for stems greater than 5 cm diameter, adjusted for nested sampling with fpc                              | t DM ha <sup>-1</sup>           | 42.6    |
| agb_ge10_ha      | num   | Aboveground woody biomass per hectare for stems greater than 10 cm diameter, adjusted for nested sampling with fpc                             | t DM ha <sup>-1</sup>           | 42.6    |
| agb_ge20_ha      | num   | Aboveground woody biomass per hectare for stems greater than 20 cm diameter, adjusted for nested sampling with fpc                             | t DM ha <sup>-1</sup>           | 42.6    |
| diam_scale       | num   | Diameter distribution scaling parameter. The slope of a linear model of log-transformed diameter class frequency values in 5 cm diameter bins. | NA                              | 1.55    |
| diam_scale_se    | num   | Standard error on the slope parameter of the diameter distribution scaling parameter.                                                          | NA                              | 0.544   |
| weibull_scale    | num   | Scale parameter of a two-parameter diameter Weibull probability distribution.                                                                  | NA                              | 1.47    |
| weibull_shape    | num   | Shape parameter of a two-parameter diameter Weibull probability distribution.                                                                  | NA                              | 0.63    |

Table 1: Column names for plots table. *(continued)*

| Column           | Class | Description                                                                                                                                                                                                | Units                           | Example |
|------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------|
| weibull_scale_se | num   | Standard error on the scale parameter of the diameter class distribution                                                                                                                                   | NA                              | 0.444   |
| weibull_shape_se | num   | Standard error on the shape parameter of the diameter class distribution                                                                                                                                   | NA                              | 0.135   |
| baobab           | log   | Do members of the Adansonia genus comprise > 10% of the aboveground biomass? This column is included because Adansonia spp. peculiar growth form can drastically inflate the aboveground biomass of a plot | NA                              | TRUE    |
| stocking         | num   | Number of stems per plot adjusted for nested sampling with fpc                                                                                                                                             | n                               | 54.23   |
| stocking_ge5     | num   | Number of stems greater than 5 cm diameter per plot, adjusted for nested sampling with fpc                                                                                                                 | n                               | 43.2    |
| stocking_ge10    | num   | Number of stems greater than 10 cm diameter per plot, adjusted for nested sampling with fpc                                                                                                                | n                               | 34.5    |
| stocking_ge20    | num   | Number of stems greater than 20 cm diameter per plot, adjusted for nested sampling with fpc                                                                                                                | n                               | 3.24    |
| ba               | num   | Basal area per plot, adjusted for nested sampling with fpc                                                                                                                                                 | m <sup>2</sup> ha <sup>-1</sup> | 3.2     |
| agb              | num   | Aboveground woody biomass per plot, adjusted for nested sampling with fpc                                                                                                                                  | t DM ha <sup>-1</sup>           | 5.4     |
| agb_ge5          | num   | Aboveground woody biomass per plot for stems greater than 5 cm diameter, adjusted for nested sampling with fpc                                                                                             | t DM ha <sup>-1</sup>           | 42.6    |
| agb_ge10         | num   | Aboveground woody biomass per plot for stems greater than 10 cm diameter, adjusted for nested sampling with fpc                                                                                            | t DM ha <sup>-1</sup>           | 42.6    |
| agb_ge20         | num   | Aboveground woody biomass per plot for stems greater than 20 cm diameter, adjusted for nested sampling with fpc                                                                                            | t DM ha <sup>-1</sup>           | 42.6    |

Table 2: Column names for stems table.

| Column             | Class | Description                                                                                                                                                                                                                                                                                                                                                                          | Units           | Example                                                          |
|--------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------|
| plot_id            | chr   | Unique identifier for plots. A three letter code, followed by a number referring to the particular plot. Three letter codes are normally, but not always, created like this: Angola (A), in Bicuar National Park (B), by John Godlee (G). The three letter code should be conserved within a dataset, i.e. a group of plots owned by a particular researcher in a particular locale. | NA              | ABG_1                                                            |
| subplot_id         | chr   | If the plot is divided into multiple parcels, but not "nested", i.e. sampling effort is equal across all parcels, the parcel within which the stem is found.                                                                                                                                                                                                                         | NA              | A                                                                |
| tree_id            | chr   | Unique identifier for trees within a plot, normally the first stem ID within the tree.                                                                                                                                                                                                                                                                                               | NA              | 45                                                               |
| stem_id            | chr   | Unique identifier for stems within a plot. Normally the tag ID if the plot has tags.                                                                                                                                                                                                                                                                                                 | NA              | 6                                                                |
| tag_id             | chr   | The code of the tag attached to the tree or stem in the field. If tags were not used, leave this column blank.                                                                                                                                                                                                                                                                       | NA              | BP094                                                            |
| census_date        | chr   | The date of the census. This can be either YYYY, YYYY-MM, or YYYY-MM-DD.                                                                                                                                                                                                                                                                                                             | NA              | 26/06/2022                                                       |
| species_orig_binom | chr   | The binomial name of the tree species, as provided by the data originator                                                                                                                                                                                                                                                                                                            | NA              | Burkea africana                                                  |
| species_orig_local | chr   | The local name of the tree species/morphospecies, as provided by the data originator                                                                                                                                                                                                                                                                                                 | NA              | mumwei                                                           |
| notes_stem         | chr   | Miscellaneous notes related to the tree or stem                                                                                                                                                                                                                                                                                                                                      | NA              | Leaning around 45 degrees and with a beehive in the fluted trunk |
| diam               | num   | Stem diameter measurement in cm                                                                                                                                                                                                                                                                                                                                                      | cm              | 6.4                                                              |
| pom                | num   | The height at which diam was taken, in metres to one decimal place (nearest 10 cm)                                                                                                                                                                                                                                                                                                   | m               | 1.3                                                              |
| x_grid             | num   | The x coordinate of the stem within the plot if a grid method was used. In metres to nearest 10 cm                                                                                                                                                                                                                                                                                   | m               | 54.2                                                             |
| y_grid             | num   | The y coordinate of the stem within the plot if a grid method was used. In metres to nearest 10 cm                                                                                                                                                                                                                                                                                   | m               | 35.6                                                             |
| latitude           | num   | The latitude coordinate of the stem within the plot if a GPS method was used. WGS84 EPSG4326                                                                                                                                                                                                                                                                                         | decimal degrees | -50.425622                                                       |
| longitude          | num   | The longitude coordinate of the stem within the plot if a GPS method was used. WGS84 EPSG4326                                                                                                                                                                                                                                                                                        | decimal degrees | 30.446453                                                        |

Table 2: Column names for stems table. (*continued*)

| Column            | Class | Description                                                                                                                                                                                                                                                                     | Units           | Example |
|-------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|
| angle             | num   | If 'plot_shape' == "circle", and angular stem coordinates were used, the angle (0 = North) direction of the stem from the plot centre, in degrees (0-360)                                                                                                                       | degrees         | 43.2    |
| distance          | num   | If 'plot_shape' == "circle", and angular stem coordinates were used, the distance of the stem from the plot centre. In metres to the nearest 10 cm                                                                                                                              | m               | 3.5     |
| height            | num   | The height of the tree, in metres to one decimal place (nearest 10 cm)                                                                                                                                                                                                          | m               | 3.5     |
| bole_height       | num   | The height from the ground to the base of the lowest branch also called 'clear length'                                                                                                                                                                                          | m               | 6.3     |
| crown_x           | num   | If two crown diameters are taken perpendicular to one another, the first of these measurements. If one crown diameter was taken, that crown diameter, in metres to one decimal place (nearest 10 cm).                                                                           | m               | 5.2     |
| crown_y           | num   | If two crown diameters are taken perpendicular to one another, the second of these measurements. If one crown diameter was taken, this remains blank, in metres to one decimal place (nearest 10 cm).                                                                           | m               | 3.5     |
| stump_diam_top    | num   | If the stem is a stump, the diameter at the top of the stump, in cm.                                                                                                                                                                                                            | cm              | 10.4    |
| stump_diam_base   | num   | If the stem is a stump, the diameter at the base of the stump (30 cm), in cm.                                                                                                                                                                                                   | cm              | 12.5    |
| stem_status       | fct   | What is the growth status of the stem. Possible values are: "r" if the stem is topkilled and resprouting, "d" if the stem and the rest of the tree is completely dead (i.e. base appears dead and all stems are dead), "a" if the stem is alive with no abnormal growth status. | NA              | r       |
| broken_per_remain | num   | If the stem is broken, approximately how much is remaining? As a decimal, e.g. 60% of stem remaining = 0.6, to one decimal place (nearest 10%)                                                                                                                                  | proportion      | 0.2     |
| broken_height     | num   | If the stem is broken, at what height is it broken. In metres to one decimal place (nearest 10 cm)                                                                                                                                                                              | m               | 4.5     |
| wound_area        | num   | If the stem is wounded, the area of the wound.                                                                                                                                                                                                                                  | cm <sup>2</sup> | 6       |

Table 2: Column names for stems table. (*continued*)

| Column             | Class | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Units      | Example                                      |
|--------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------|
| stem_mode          | fct   | Various flags denoting stem mode. Possible values are: "b" - broken (material above break is detached), "c" - has climbing or strangling or liana plants in canopy, "f" - fallen, "g" - hollow, "j" - bark peeling or lost, "k" - stem has bracket fungi, "l" - leaning (>15deg), "m" - on a termite mound, "n" - a new recruit, "p" - snapped (material above break is attached), "q" - the stem location is unknown and was not found (contrast with "v"), "s" - standing, "t" - stump i.e. broken below POM, "u" - uprooted, "v" - vanished (location of tree found but stem missing), "w" - stem is a liana, "y" - wounded, "z" - sick (i.e. lost most leaves, discoloured foliage, drooping branches) or rotten wood | NA         | P                                            |
| damage_cause       | fct   | If the stem is damaged, what caused it? Possible values are: "a" if animals (not elephants, not insects), "e" if elephants, "f" if fire, "h" if humans (e.g. cut, ringbarked, broken or beehive), "i" if frost, "l" if lightning, "m" if termites, "n" if neighbouring tree, "o" if other (describe in damage_cause_notes), "q" if unknown, "w" if wind                                                                                                                                                                                                                                                                                                                                                                   | NA         | N                                            |
| damage_cause_notes | chr   | If damage_cause = "o", describe the specific method of damage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NA         | chainsaw marks and machete on small branches |
| cambium            | num   | The proportion of trunk surface area with exposed vascular cambium, between the ground and 2 m.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | proportion | 0.22                                         |
| decay              | num   | What level of decay is the stem? Scale runs from 1 (no decay) to 5 (very decayed). Follow the "Decay class" table in the SEOSAW protocol. To 0.5 of a decay level (e.g. 1.5, 1, 3.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | NA         | 4.5                                          |
| diam_method        | fct   | What method was used to derive the stem diameter measurement? Possible values are: "dbh_tape" if a tape measure with a diameter conversion scale was used, "tape" if a normal tape measure was used and the circumference converted to diameter, "caliper_1" if a single caliper measurement was used, "caliper_2" if two caliper measurements at 90 degrees with the mean taken, "other".                                                                                                                                                                                                                                                                                                                                | NA         | dbh_tape                                     |

Table 2: Column names for stems table. *(continued)*

| Column             | Class | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Units              | Example                                     |
|--------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------|
| height_method      | fct   | What method was used to derive the height measurement? Possible values are: "eye" if estimated by eye, "rangefinder" if estimated using laser or ultrasonic apparatus with an electronic tilt sensor e.g. Nikon Forestry Pro, "clinometer" if estimated using a mechanical Clinometer, "hyposometer" if estimated using a laser vertex hypsometer from below the tree using last return, "direct" if estimated by climbing the tree or adjacent tower and measuring the height with tape or stick, "other" | NA                 | eye                                         |
| voucher_id         | chr   | If a herbarium voucher has been collected, what is the code to identify it? 1, Wedza_01, MHR-04b are all valid. No spaces please.                                                                                                                                                                                                                                                                                                                                                                          | NA                 | KEW_FK_42                                   |
| next_census_check  | chr   | If any measured values seem wrong or, note checks to be performed during the next census.                                                                                                                                                                                                                                                                                                                                                                                                                  | NA                 | Appears to have resurrected check mortality |
| fpc                | num   | Fractional Probability of Inclusion. If plots are nested with varying sampling effort within plots for stems of different size, the probability of a stem this size occurring in the whole plot.                                                                                                                                                                                                                                                                                                           | proportion         | 0.35                                        |
| crown_area         | num   | Area of the tree crown, measured as an ellipse using either one or two crown diameter measurements, if present. In m <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                          | m <sup>2</sup>     | 4.5                                         |
| agb                | num   | Aboveground biomass of stem, in tonnes of dry matter                                                                                                                                                                                                                                                                                                                                                                                                                                                       | t DM               | 0.093                                       |
| wood_density_mean  | num   | Mean estimated wood density of stem, in g cm <sup>-3</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | g cm <sup>-3</sup> | 0.042                                       |
| wood_density_sd    | num   | Standard deviation of estimated wood density of stem, in g cm <sup>-3</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                | g cm <sup>-3</sup> | 0.0004                                      |
| wood_density_lev   | chr   | Level at which wood density was estimated                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NA                 | species                                     |
| wood_density_n     | num   | Number of individuals used to estimate wood density                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NA                 | 613                                         |
| ba                 | num   | Basal area of stem, in m <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | m <sup>2</sup>     | 0.46                                        |
| diam_adj           | num   | Adjusted stem diameter at 1.3 m height accounting for tree taper                                                                                                                                                                                                                                                                                                                                                                                                                                           | cm                 | 5.3                                         |
| diam_inc           | num   | Diameter growth increment between current census and previous census                                                                                                                                                                                                                                                                                                                                                                                                                                       | cm                 | 5.4                                         |
| diam_gr            | num   | Diameter annual growth rate between current census and previous census                                                                                                                                                                                                                                                                                                                                                                                                                                     | cm y <sup>-1</sup> | 0.23                                        |
| census_int         | num   | Census interval, in decimal years, between current census and previous census                                                                                                                                                                                                                                                                                                                                                                                                                              | years              | 3.443                                       |
| species_name_sanit | chr   | Species name after automated taxonomy cleaning by sanitizeTaxa()                                                                                                                                                                                                                                                                                                                                                                                                                                           | NA                 | Protea cf. petiolaris                       |
| species_name_clean | chr   | Species name after taxize checks and synonymy corrections                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NA                 | Burkea africana                             |
| genus_clean        | chr   | Cleaned genus name, taken from 'species_name_clean'                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NA                 | Burkea                                      |
| species_clean      | chr   | Cleaned species epithet, taken from 'species_name_clean'                                                                                                                                                                                                                                                                                                                                                                                                                                                   | NA                 | africana                                    |
| subspecies_clean   | chr   | Cleaned subspecies epithet, taken from 'species_name_clean'                                                                                                                                                                                                                                                                                                                                                                                                                                                | NA                 | africana                                    |

Table 2: Column names for stems table. (*continued*)

| Column           | Class | Description                                                                                                                    | Units | Example                                                                                      |
|------------------|-------|--------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------|
| confer_clean     | chr   | Cleaned cf. part of species epithet, taken from<br>'species_name_clean'                                                        | NA    | polyantha                                                                                    |
| variety_clean    | chr   | Cleaned var. part of species epithet, taken from<br>'species_name_clean'                                                       | NA    | micrantha                                                                                    |
| family_gnr       | chr   | Matched family name using taxize::gnr_resolve()                                                                                | NA    | Fabaceae                                                                                     |
| species_name_gnr | chr   | Name matched during the GNR resolve                                                                                            | NA    | Burkea africana                                                                              |
| taxon_rank_gnr   | chr   | All rank names of classification from GNR resolve, separated by " "                                                            | NA    | superkingdom kingdom phylum subphylum                                                        |
| taxon_gnr        | chr   | All ranks of classification from GNR resolve, separated by " "                                                                 | NA    | Eukaryota Viridiplantae Streptophyta Stratiotales Burkeales Burkeaceae Burkea<br>makondensis |
| base_rank_gnr    | chr   | The lowest rank identified in GNR resolve                                                                                      | NA    | species                                                                                      |
| species_name_cjb | chr   | Species name matched with the CJB African Plants database                                                                      | NA    | Burkea africana                                                                              |
| subdiv_id        | chr   | If the plot has been divided into smaller 1 ha subdivisions, the<br>names of 1 ha plot subdivisions to which the stem belongs. | NA    | ABG_1_S4                                                                                     |

Table 3: Column names for small stems table.

| Column             | Class | Description                                                                                                                                                                                                                                                                                                                                                                                           | Units | Example                     |
|--------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------|
| plot_id            | chr   | Standardised method of identifying plots. A three letter code, followed by a number referring to the particular plot. Three letter codes are normally, but not always, created like this: Angola (A), in Bicuar National Park (B), by John Godlee (G). The three letter code should be conserved within a dataset, i.e. a group of plots owned by a particular researcher within a particular locale. | NA    | SSM_1                       |
| census_date        | chr   | The date the measurement was taken. This can be formatted as either YYYY, YYYY-MM, or YYYY-MM-DD                                                                                                                                                                                                                                                                                                      | NA    | 05/05/2022                  |
| transect_id        | chr   | ID of transect                                                                                                                                                                                                                                                                                                                                                                                        | NA    | T1                          |
| transect_length    | num   | Length of transects, in m to nearest 10 cm. If ‘transect_id’ is not provided, the total length of all transects in the plot.                                                                                                                                                                                                                                                                          | m     | 100                         |
| transect_width     | num   | Width of belt transect, in m to nearest 10 cm                                                                                                                                                                                                                                                                                                                                                         | m     | 2                           |
| species_orig_binom | chr   | The binomial name of the tree species, as provided by the data originator                                                                                                                                                                                                                                                                                                                             | NA    | Burkea africana             |
| species_orig_local | chr   | The local name of the tree species/morphospecies, as provided by the data originator                                                                                                                                                                                                                                                                                                                  | NA    | mumwei                      |
| n_alive            | int   | Number of living stems found for a given species                                                                                                                                                                                                                                                                                                                                                      | NA    | 50                          |
| n_dead             | int   | Number of dead stems found for a given species                                                                                                                                                                                                                                                                                                                                                        | NA    | 4                           |
| notes_small_stems  | chr   | Miscellaneous notes on each species of small stem                                                                                                                                                                                                                                                                                                                                                     | NA    | Could be Combretum collinum |

Table 4: Column names for coarse woody debris table.

| Column             | Class | Description                                                                                                                                                                                                                                                                                                                                                                                           | Units | Example                                        |
|--------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------|
| plot_id            | chr   | Standardised method of identifying plots. A three letter code, followed by a number referring to the particular plot. Three letter codes are normally, but not always, created like this: Angola (A), in Bicuar National Park (B), by John Godlee (G). The three letter code should be conserved within a dataset, i.e. a group of plots owned by a particular researcher within a particular locale. | NA    | SSM_1                                          |
| census_date        | chr   | The date the measurement was taken. This can be formatted as either YYYY, YYYY-MM, or YYYY-MM-DD                                                                                                                                                                                                                                                                                                      | NA    | 05/05/2022                                     |
| transect_id        | chr   | ID of transect                                                                                                                                                                                                                                                                                                                                                                                        | NA    | NW                                             |
| transect_length    | num   | Length of transects, in m to nearest 10 cm. If 'transect_id' is not provided, the total length of all transects in the plot.                                                                                                                                                                                                                                                                          | m     | 100                                            |
| piece_id           | chr   | ID of piece of coarse woody debris. A piece is a single contiguous bit.                                                                                                                                                                                                                                                                                                                               | NA    | 1                                              |
| section_id         | chr   | ID of section of coarse woody debris. A section is part of a piece.                                                                                                                                                                                                                                                                                                                                   | NA    | BP094                                          |
| tag_id             | chr   | The code of any tags attached to the piece in the field.                                                                                                                                                                                                                                                                                                                                              | NA    | BP094                                          |
| species_orig_binom | chr   | The binomial name of the tree species, as provided by the data originator                                                                                                                                                                                                                                                                                                                             | NA    | Burkea africana                                |
| species_orig_local | chr   | The local name of the tree species/morphospecies, as provided by the data originator                                                                                                                                                                                                                                                                                                                  | NA    | mumwei                                         |
| section_diam       | num   | Diameter of section of CWD at point of intersection with transect, in cm.                                                                                                                                                                                                                                                                                                                             | cm    | 6.8                                            |
| section_length     | num   | Length of section of CWD, in m to nearest cm                                                                                                                                                                                                                                                                                                                                                          | m     | 4.5                                            |
| decay              | num   | What level of decay is the stump? Scale runs from 1 (no decay) to 5 (very decayed). Follow the "Decay class" table in the SEOSAW protocol. To 0.5 of a decay level (e.g. 1.5, 1, 3.5)                                                                                                                                                                                                                 | NA    | 4.5                                            |
| notes_cwd          | chr   | Miscellaneous notes related to the piece of CWD                                                                                                                                                                                                                                                                                                                                                       | NA    | Approximately half of the trunk has broken off |

Table 5: Column names for ground layer biomass table.

| Column              | Class | Description                                                                                                                                                                                                                                                                                                                                                                                            | Units                          | Example                |
|---------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------|
| plot_id             | chr   | Standardised method of identifying plots. A three letter code, followed by a number referring to the particular plot. Three letter codes are normally, but not always, created like this: Angola (A), in Bicular National Park (B), by John Godlee (G). The three letter code should be conserved within a dataset, i.e. a group of plots owned by a particular researcher within a particular locale. | NA                             | SSM_1                  |
| census_date         | chr   | The date the measurement was taken. This can be formatted as either YYYY, YYYY-MM, or YYYY-MM-DD                                                                                                                                                                                                                                                                                                       | NA                             | 11/06/2022             |
| quadrat_id          | chr   | Unique identifier for quadrats within a plot                                                                                                                                                                                                                                                                                                                                                           | NA                             | Q1.1                   |
| quadrat_area        | num   | Area covered by the quadrat.                                                                                                                                                                                                                                                                                                                                                                           | m <sup>2</sup>                 | 0.25                   |
| sward_height        | num   | Height of the grass sward to nearest cm                                                                                                                                                                                                                                                                                                                                                                | cm                             | 25                     |
| bare_cover          | num   | Proportional bare ground cover. Bare ground has no litter, just bare soil.                                                                                                                                                                                                                                                                                                                             | proportion                     | 0.45                   |
| rock_cover          | num   | Proportional bare rock cover.                                                                                                                                                                                                                                                                                                                                                                          | proportion                     | 0.73                   |
| dung                | log   | Is there animal dung larger than a human fist inside the quadrat?                                                                                                                                                                                                                                                                                                                                      | NA                             | TRUE                   |
| fire                | log   | Is there evidence of a previous fire inside the quadrat, e.g. charcoal.                                                                                                                                                                                                                                                                                                                                | NA                             | FALSE                  |
| under_canopy        | log   | Is the quadrat directly under a tree canopy?                                                                                                                                                                                                                                                                                                                                                           | NA                             | TRUE                   |
| canopy_cover        | prop  | The proportion of the ground (periscope, aerial photo, canopy dimensions), or sky hemisphere (hemi-photo, mirror densitometer), covered by a tree canopy. Measured as a proportion between 0 and 1, e.g. 60% = 0.6                                                                                                                                                                                     | proportion                     | 0.28                   |
| canopy_cover_method | fct   | Possible values are: "periscope_densitometer", "mirror_densitometer", "hemispherical_photo", "aerial_photo", "canopy_dimensions", "other"                                                                                                                                                                                                                                                              | NA                             | periscope_densitometer |
| lai_above           | num   | Leaf Area Index above the grass sward, measured with an AccuPAR Ceptometer.                                                                                                                                                                                                                                                                                                                            | m <sup>2</sup> m <sup>-2</sup> | 1.72                   |
| lai_below           | num   | Leaf Area Index below the grass sward, measured with an AccuPAR Ceptometer.                                                                                                                                                                                                                                                                                                                            | m <sup>2</sup> m <sup>-2</sup> | 4.55                   |
| litter_depth        | num   | Depth of leaf litter inside the quadrat, to nearest cm.                                                                                                                                                                                                                                                                                                                                                | cm                             | 12                     |
| grass_wet_mass      | num   | The wet mass of grass material collected inside the quadrat.                                                                                                                                                                                                                                                                                                                                           | g                              | 52.5                   |
| other_leaf_wet_mass | num   | The wet mass of non-grass leaf material collected inside the quadrat.                                                                                                                                                                                                                                                                                                                                  | g                              | 25.6                   |
| wood_wet_mass       | num   | The wet mass of woody litter collected inside the quadrat.                                                                                                                                                                                                                                                                                                                                             | g                              | 42.8                   |
| litter_wet_mass     | num   | The wet mass of leaf litter collected inside the quadrat.                                                                                                                                                                                                                                                                                                                                              | g                              | 24.7                   |
| succ_wet_mass       | num   | The wet mass of succulent material collected inside the quadrat.                                                                                                                                                                                                                                                                                                                                       | g                              | 35.6                   |
| grass_dry_mass      | num   | The dry mass of grass material collected inside the quadrat.                                                                                                                                                                                                                                                                                                                                           | g                              | 5.2                    |
| other_leaf_dry_mass | num   | The dry mass of non-grass leaf material collected inside the quadrat.                                                                                                                                                                                                                                                                                                                                  | g                              | 2.5                    |
| wood_dry_mass       | num   | The dry mass of woody litter collected inside the quadrat.                                                                                                                                                                                                                                                                                                                                             | g                              | 4.2                    |

Table 5: Column names for ground layer biomass table. (*continued*)

| Column              | Class | Description                                                      | Units | Example                     |
|---------------------|-------|------------------------------------------------------------------|-------|-----------------------------|
| litter_dry_mass     | num   | The dry mass of leaf litter collected inside the quadrat.        | g     | 2.4                         |
| succ_dry_mass       | num   | The dry mass of succulent material collected inside the quadrat. | g     | 3.5                         |
| notes_quadrat       | chr   | Miscellaneous notes on each quadrat                              | NA    | Pig diggings inside quadrat |
| grass_moisture      | num   | Moisture content of grass                                        | g     | 3.6                         |
| other_leaf_moisture | num   | Moisture content of other leaves                                 | g     | 2.8                         |
| wood_moisture       | num   | Moisture content of woody litter                                 | g     | 2.6                         |
| litter_moisture     | num   | Moisture content of leaf litter                                  | g     | 7.4                         |
| succ_moisture       | num   | Moisture content of succulent material                           | g     | 7.3                         |

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